

Base-R Exercise 2 - SAS Cars Merge Aggregate

Excercise 2

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Steps -----

- 1) Open SAS and save the preloaded “Cars” data file to your computer (e.g. downloads, desktop). Leave the file in SAS format (.SAS7BDAT).

See file in explorer!

- 2) Open R and install the “haven” package, then use the library command to load the package into the R software.

install.packages("haven") or use package manager to the right like shown in class

Save the sashelp.cars data set as variable
`library(haven)`

- 3) Import the SAS “Cars” file using the “haven” package.

```
cars_orig <- read_sas("C:/Users/aiden/OneDrive - University of Toledo/Spring 2026 UT NOW/Data Science I
```

- 4) Make a copy of cars, named “cars2”

```
cars2 <- cars_orig
```

- 5) Sort cars2 by make and model.

```
cars2 <- cars2[order(cars2$Make, cars2$Model),]
```

- 6) Create a numeric ID variable (using any method).

```
cars2$id <- 0  
cars2$id <- as.integer(rownames(cars2))
```

- 7) Convert all variable names to lower case.

```
names(cars2) <- tolower(names(cars2))
```

8) Create a dataframe called cars_half1 that keeps only the variables: id, make, mpg_city, mpg_highway

```
cars_half1 <- subset(cars2, select = c(id, make, mpg_city, mpg_highway))
```

9) Create a dataframe called cars_half2 that keeps only the variables: id, model, type, msrp, weight

```
cars_half2 <- subset(cars2, select = c(id, model, type, msrp, weight))
```

10) Merge cars_half1 and cars_half2 by id (name the merged file "cars_merged")

```
cars_merged <- merge(cars_half1, cars_half2, by = "id")
```

11) Using cars_merged, create a new variable called "mpg_avg" that averages mpg_city and mpg_highway.

```
cars_merged$mpg_avg <- (cars_merged$mpg_city + cars_merged$mpg_highway) / 2
```

12) Use the "table" function to obtain a frequency count of cars by "type" (write the counts by type using comments).

```
table(cars_merged$type)
```

```
##
## Hybrid  Sedan Sports    SUV  Truck  Wagon
##      3    262    49     60    24    30
```

```
# Hybrid: 3
# Sedan: 262
# Sports: 49
# SUV: 60
# Truck: 24
# Wagon: 30
```

13) Remove all cars that are type "Hybrid" by using a filter variable and bracket notation (two step process).

```
hybrid_check <- cars_merged$type == "Hybrid" # gotta remember that comparison operator instead of equals
cars_merged <- cars_merged[!hybrid_check, ]
```

14) Aggregate all vehicles by type to identify the mean of mpg_avg for each type.

```
type_means <- aggregate(mpg_avg ~ type, data = cars_merged, mean)
type_means
```

```
##      type  mpg_avg
## 1  Sedan 24.85687
## 2 Sports 21.94898
## 3   SUV 18.30000
## 4  Truck 18.75000
## 5  Wagon 24.50000
```

15) Write a comment that lists the mean values for each type of car.

```
# type    mpg_avg
# Sedan  24.85687
# Sports  21.94898
# SUV     18.30000
# Truck   18.75000
# Wagon   24.50000
```